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Retrieving similar cases for construction project risk management using Natural Language Processing techniques

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ABSTRACT

Case-based reasoning (CBR) is an important approach in construction project risk management. It emphasises that previous knowledge and experience of accidents and risks are highly valuable and could contribute to avoiding similar risks in new situations. In the CBR cycle, retrieving useful information is the first and the most important step. To facilitate the CBR for practical use, some researchers and organisations have established construction accident databases and their size is growing. However, as those documents are written in everyday language using different ways of expression, how information in similar cases is retrieved quickly and accurately from the database is still a huge challenge. In order to improve the efficiency and performance of risk case retrieval, this paper proposes an approach of combining the use of two Natural Language Processing (NLP) techniques, i.e. Vector Space Model (VSM) and semantic query expansion, and outlines a framework for this Risk Case Retrieval System. A prototype system is developed using the Python programming language to support the implementation of the proposed method. Preliminary test results show that the proposed system is capable of retrieving similar cases automatically and returning, for example, the top 10 similar cases.

1. Introduction

Construction is among the most hazardous and dangerous industries in the world [1]. In the U.S., it is reported that over 157 bridges collapsed between 1989 and 2000 [2], and > 26,000 workers lost their lives on construction sites during the past two decades [3]. Globally, the International Labour Organization (ILO) estimates that approximately 60,000 fatal accidents happen every year [4]. Such serious accidents may not only lead to a bad reputation for the construction industry but also trigger further risks such as project failure, financial difficulty and time overruns. To avoid such serious accidents and improve the performance of risk management in future projects, a few studies [5,6] suggested project practitioners should learn the valuable lessons from previous accidents and embed the consideration of risk management into the development process of a project. Learning from the past is a fundamental process in project risk management that helps individuals and organisations understand when, what and why incidents happened, and how to avoid repeating past mistakes [7].

In general, the process of solving new problems based on experience of similar past problems is known as Case-Based Reasoning (CBR) [8], which examines what has taken place in the past and applies it to a new situation [9], and could be of particular help in identifying and mitigating project risks at early stages, e.g. design and construction planning. In order to facilitate CBR for practical use in the construction industry, some efforts have been observed in collecting risk cases and establishing a risk case database. For example, Zhang et al. [10] developed a database containing 249 incident cases to support risk management for metro operations in Shanghai. And there are > 600 verified reports about structural risks on the Structural-Safety website [11] and similarly the National Institute for Occupational Safety and Health (NIOSH) [12] has established a database of over 249 reports on construction accidents. In addition, for identifying the reasons that contribute to collision injuries, Esmaeili and Hallowell [13] reviewed and analysed over 300 accident reports. However, as a risk case database often contains a huge amount of data where reports are written in everyday language, manually reviewing, analysing and understanding these reports is a time-consuming, labour-intensive and inefficient work. Failure in extracting 'correct' cases and information within a limited time often may mean that the importance of learning from past experience is missed. Hence, some researchers [7,14,15] pointed out that a key challenge in current CBR research for project risk management is how to quickly and accurately retrieve relevant risk case data from the database so that knowledge and experience could be incorporated into new risk identification and assessment in a timely manner.

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In recent years, with the development and growing use of Natural Language Processing (NLP) in the computer science discipline, some researchers have been trying to introduce NLP into the construction industry to address the analysis and management issues of textual documents, e.g. retrieval of CAD drawings [16], automatic analysis of injury reports [14], and automatic clustering of construction project documents based on textual similarity [17]. It could be seen that NLP is a promising technique in assisting the knowledge and case retrieval of CBR. However, very few studies have been found in this field. In addition, Goh and Chua [7] stated that very few NLP tools nowadays appear to be suitable for the construction industry.

In order to improve the efficiency and performance of risk case retrieval, this paper proposes an approach of combining the use of two NLP techniques, i.e. Vector Space Model (VSM) and semantic query expansion, and outlines a framework for the Risk Case Retrieval System. A prototype system is developed with the Python programming language to support the implementation of the proposed method.

The rest of this paper is organised as follows. Section 2 introduces the background and current challenges of CBR in project risk management, and discusses the potential of integrating NLP in CBR and the motivation of this study. The system architecture and methodologies used in this study are described in Section 3. In Section 4, a prototype system is developed with Python. A simple example is used for illustrating the proposed method, and a preliminary test is conducted to evaluate the system. Finally, the implications, limitations, recommendations for future research and conclusions are addressed in Sections 5 and 6.

2. Background and point of departure

2.1. Current challenges in case retrieval

CBR is a branch of Artificial Intelligence (AI) and its origin can be traced back to the work of Roger Schank and his students in the early 1980s [15,18,19]. The core philosophy behind CBR is that previous knowledge and experience can be recalled and used as a starting point to solve new problems in many fields. In the project management domain, CBR has been recognised as an important technique for risk identification and analysis [20] and a number of applications have been developed, e.g. construction hazard identification [7,21], safety risk analysis in subway operations [22], and construction supply chain risk management [23]. Fig. 1 shows the classical model of a CBR system adapted from a previous research by Aamodt and Plaza [24]. Basically the implementation cycle of CBR contains four main processes: RETRIEVE, REUSE, REVISE, and RETAIN (known as 'the four REs'), where RETRIEVE is the first and the most important process in any CBR systems [22].

RETRIEVE is a process of searching and determining the most similar and relevant case or cases [15,24], and its importance can be viewed from the following three main aspects: (1) it acts as the only medium for helping individuals extract information from a risk case database; (2) as a risk case database often contains a large number of 'human language' based documents, the performance of case retrieval will have direct influence on the quality and accuracy of retrieved cases; and (3) the inefficiency of case retrieval seriously affects the user experience, which may lead to the importance of previous knowledge and experience being overlooked.

Currently scoring the similarity through allocating weights to factors is the most common method in case retrieval. For example, Lu et al. [22] employed a semantic network approach to calculate the similarity value between two accident precursors. Karim and Adeli [25] collected risk data into Excel tables and developed an attribute based schema for calculating the similarity between two cases. Goh and Chua [7] proposed a sub-concept approach based on a semantic network. Other efforts include, for example, evaluation of attributes [9], taxonomy tree approach [26], ontology-based method [27].

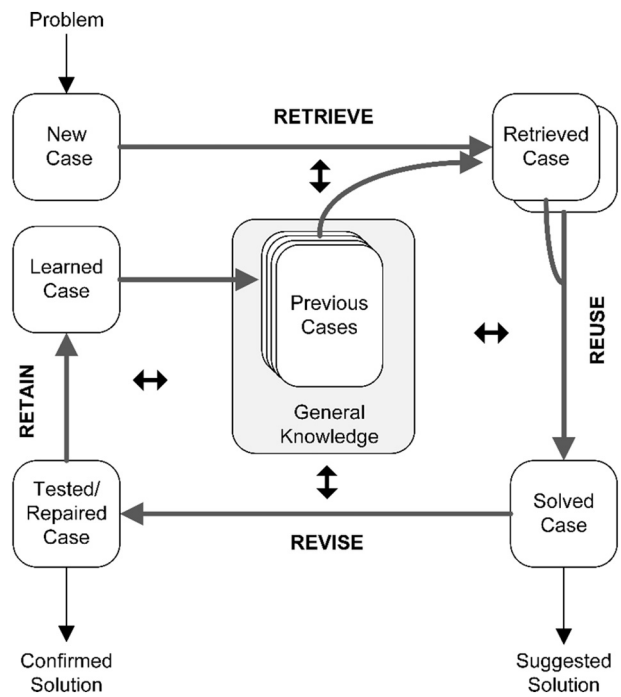

Fig. 1. Classical model of a CBR system [24].

However, challenges and limitations also exist in current efforts, which are summarised as follows:

(1) Existing studies are very limited in scope. For example, the CBR system developed by Lu et al. [22] predefined the potential accidents in subway operations and the similarity calculation is based on attributes that are to some extent subjective. Similarly, the prototype proposed by Karim and Adeli [25] calculated the similarity index based on different weights of attributes and is only designed for highway work zone traffic management.
(2) A large amount of pre-processing or preparation work is needed. For instance, the sub-concept approach [7] needs to establish a semantic network map of variables and each semantic network is constructed based on analysis of cases and allocation of weights. Goh and Chua [7] acknowledged that organisations implementing the system need to consider the cost for establishing and maintaining the semantic networks and risk cases.
(3) Very few studies have been found in addressing the challenge of semantic similarity in case retrieval. Semantic similarity is defined as "*a metric defined over a set of terms or documents, where the idea of distance between them is based on the likeness of their meaning or semantic content as opposed to similarity which can be estimated regarding their syntactical representation*" [28]. Semantic similarity problems can be observed in, for example, synonyms (e.g. 'building' and 'house'), hyponyms (e.g. 'structure' and 'bridge'), and even related words (e.g. 'car' and 'bus'). Because risk case reports are all written in everyday human language and in different ways of expressing meaning by different individuals or organisations, the outcomes of case retrieval will be incomplete if a CBR system fails to consider semantic similarity. Therefore, Mantaras et al. [15] pointed out that improving the performance through more effective approaches to similarity assessment has been an important research focus in CBR.

2.2. Natural Language Processing

Natural Language Processing (NLP) is an interdisciplinary topic overlapping in computational linguistics, AI, and computer science that deals with the interactions between computer and human languages

[29]. NLP started its early work in the 1950s in exploring the fully automatic translation between different languages [30], and in recent years has seen a rapid increase in use and development in computer science. The application areas of NLP are very wide including, for example, machine translation, question answering, speech recognition and information retrieval [31].

Information retrieval (IR) refers to the process and activity of extracting useful information from a collection of information resources [32]. Due to the needs of managing and using the fast-growing volume of information [33], many IR systems have been developed and the best examples include web search engines (e.g. Google and Yahoo), and library resource retrieval systems [34].

In the construction industry, even a small project generates a large amount of digital information such as specifications, computer-aided drawings, and structural analysis reports [14,35]. In addition, in order to learn from past experience and avoid similar accidents in new projects, lots of investigations and analysis on previous accidents have been conducted and the resulting reports and feedbacks are important to improving the existing knowledge and standards [36]. Currently major companies and organisations are using databases for managing those accident reports [14]. However, new documents continually need to be added into databases and therefore the size of databases is increasing. Moreover, these reports are written in human language and in different ways of expression by different individuals or organisations. As discussed in Section 2.1, a challenge is how to retrieve valuable and 'correct' information from the database quickly and efficiently.

To improve the use and management of 'human language' based engineering documents, a recent research trend is to take advantage of NLP. For example, Yu and Hsu [16] made the use of the classical VSM and developed a Content-based CAD document Retrieval System (C-CRS) for assisting the management of CAD drawings and quick retrieval of documents according to given queries. By taking the advantage of keywords extraction of NLP, Tixier et al. [14] developed a prototype supported by the R programming language for automatically extracting precursors and outcomes from unstructured injury reports. Qady and Kandil [17] proposed a method for automatic clustering of construction project documents based on textual similarity. Caldas and Soibelman [37] developed a prototype system to automatically classify a large number of electronic text documents in a hierarchical order in the information management system. Another study took the advantage of text mining and proposed an ontology-based text classification method for job hazard analysis [38]. In addition, Pereira et al. [39] presented a solution to extract valuable information from incident reports in real time to assist incident duration prediction. However, very few studies exist in this field and new investigations are still needed.

It is observed that there are two main features in applying NLP into textual document management in the construction industry:

- Firstly, most state-of-the-art studies of NLP still lie in the computer science discipline and most modern applications are often used to treat extremely large volumes of data e.g. extracting online information [40] and library management [32]. In contrast, the sizes of electronic data in any construction project and risk cases in any database are relatively small. Hence, there is a need to select the appropriate methods and techniques for specific purposes. For example, Tixier et al. [14] pointed out one difficulty in implementing machine learning for automatic safety keywords extraction is that small number of injury reports is not satisfactory as a training database and therefore they developed a NLP system based on hand-coded rules.
- Secondly, unlike online webs containing often several aspects of information, construction project data and risk cases are relatively restricted to certain topics and thus there is a need to establish the context or rules in processing them. For instance, when applying ontology and text mining into job hazard analysis, the authors predefined the list of potential safety hazards and emphasised the importance of defining the knowledge and resource scope into the construction safety domain [16].

2.3. Motivation and aim of this study

As discussed in Sections 2.1 and 2.2, some existing efforts [14,16,17] have shown that the application of NLP techniques in managing textual data is a new research trend in the construction industry and NLP has the potential to address the current challenges of case retrieval of CBR. However, very limited numbers of studies have been found in this area. In order to further improve the efficiency and performance of risk case retrieval, this paper proposes an approach of combining the use of two NLP techniques, i.e. VSM and semantic query expansion, and outlines a framework for the Risk Case Retrieval System. The idea was motivated by the following observations:

- VSM is known as one of the most important IR models [32] and it can be used for information extraction, indexing and relevancy ranking, etc. For example, Caldas and Soibelman [37] used VSM for characteristic information extraction and automatic classification of project documents. Similarly, Yu and Hsu [16] embedded VSM as a core technique in their retrieval system of CAD drawings. Hence, VSM is potentially helpful in evaluating the relevance between user need and risk cases in a CBR system.
- Understanding the relations between words (e.g. hyponymy, synonymy) is an important step in fully using the concept of semantic similarity [31]. Thus, some individuals and organisations have started to establish lexical 'dictionaries' that pre-defined the semantic relationships between words, where the most commonly used resource for English sense relations is the WordNet lexical database [31,41]. So far a number of studies [42,43] have used WordNet for improving web retrieval through expanding the query terms using related words in WordNet and have proved this approach could partially address the semantic similarity issues and improve the performance and completeness of information retrieval. Therefore, the basic principle of semantic query expansion is also applicable for improving the completeness and quality of case retrieval.

3. Framework and methodology

The overall framework and methodologies used in this study are described in this section. Specifically, the system architecture of the proposed Risk Case Retrieval System (RCRS) is presented in Section 3.1, and the three major modules of RCRS are described in detail in Sections 3.2, 3.3 and 3.4.

3.1. System architecture of the Risk Case Retrieval System

The system architecture of the proposed RCRS is illustrated in Fig. 2. The system consists of three major modules, i.e. (1) Risk case processing, (2) Query operation, and (3) Retrieval application. Firstly, the risk case processing module automatically extracts the textual information from a targeted collection of risk cases. It processes the collected textual information by a defined Sequence of Actions (SoA), i.e. tokenisation, converting all words into lowercase, lemmatisation, and removing stop words to establish a risk case content corpus. The SoA is a general approach in current NLP for processing textual documents [31]. Secondly, the query operation module reads and processes the given query by SoA. The processed query is prior scanned to match its expansion of related words in the pre-defined risk-related lexicon. The terms not found in the pre-defined risk-related lexicon are expanded by using synonyms in WordNet. Then the system scans the terms in both the original query and the expanded query, and removes those terms that do not exist in the risk case content corpus. Thirdly, the retrieval application module combines the queries and risk case corpus together and performs the query-document similarity calculations. After this, the

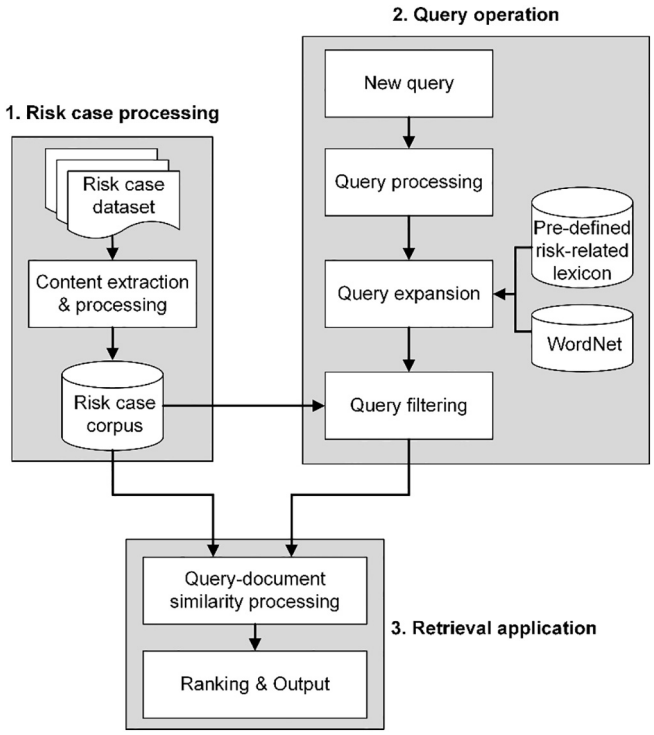

Fig. 2. System architecture of RCRS.

system ranks all documents according to their similarity scores and finally returns, for example, the top 10 documents to the users.

3.2. Risk case processing workflow

The first step in the risk case processing module is to collect risk cases through a web search method. In total 590 risk cases were collected from the following major organisational and governmental construction accident databases: (1) Structural-Safety [11], (2) the National Institute for Occupational Safety and Health (NIOSH) [12], (3) WorkSafeBC [44], (4) Occupational Safety and Health Administration [45], and (5) others (e.g. some published papers that document construction accidents). The source distribution of collected risk cases is shown in Fig. 3 and the category distribution is presented in Fig. 4. Although collecting as many risk cases as possible from every category of project risks could improve the reliability of the proposed approach, this study stopped collecting more cases due to the following reasons:

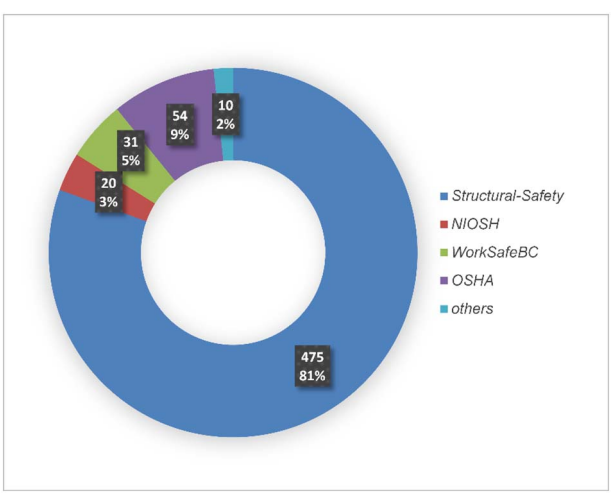

Fig. 3. Source distribution of collected risk cases.

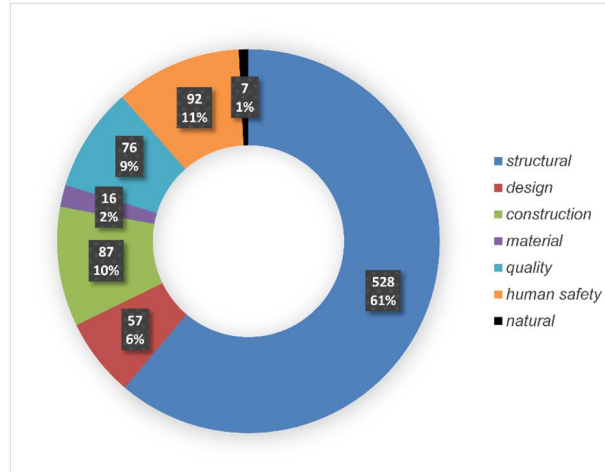

Fig. 4. Category distribution of collected risk cases.

(1) the authors have only limited research time and the main focus of this study is developing a NLP based general approach for risk case retrieval instead of establishing a complete risk case database; (2) it is observed that some risks (e.g. collapse of structure, loss of life) that may lead to severe consequences attract more attention while there are very few detailed reports available on those risks that are not so dangerous, e.g. financial loss, time overrun.

The second step is to extract the textural information from the collected reports and process them to be a risk case content corpus, which goes through the following processes:

- Tokenisation: this is a process of chopping a document up into pieces (known as 'tokens') and discarding certain characters, such as punctuation [46]. An example is illustrated in Fig. 5.
- Converting words into lowercase: this is a simple task to convert tokens into lowercase, which could improve the search results [46]. For instance, the term "Building" is converted to be "building".
- Lemmatisation: it "*usually refers to doing things properly with the use of a vocabulary and morphological analysis of words, normally aiming to remove inflectional endings only and to return the base or dictionary form of a word, which is known as the lemma*" [46]. For example, the base form "walk" may appear as "walk", "walked", "walks", or "walking" in the main text, and the process of lemmatisation is to convert those words to their base forms.
- Stop words removal: stop words are those extremely common words which have little value in helping match documents [46]. Removal of those meaningless words could largely reduce the size of collection and improve the retrieval efficiency. The stop words used in this study are presented in Table 1 which consists of two sub lists. The first list of stop words is identified by the Natural Language Toolkit (NLTK) [47], which is a suite of libraries and programs for symbolic and statistical NLP for English written in the Python programming language [48]. The second list comes from a manual selection from the top 100 words that have the most occurrences in the risk case content corpus but are identified with little value. For example, 'fig 1' has an extremely high occurrences in the whole risk case collection but its tokens (i.e. 'fig' and '1') are of little help to the

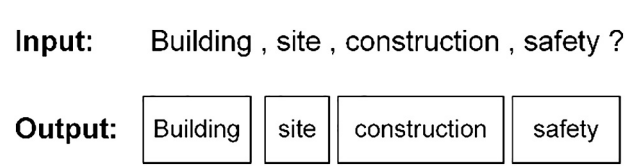

Fig. 5. An example of tokenisation.

Table 1
Stop words used in this paper.

Stop words identified by NLTK					Manually selected stop words
the	his	off	him	about	number
couldn	ain	with	doesn	re	15
shan	were	m	an	our	20
between	very	but	who	both	could
any	there	own	was	he	14
himself	while	for	during	this	16
a	hers	is	once	until	f
at	over	too	other	am	b
after	myself	just	ll	no	12
will	then	i	again	mightn	fig
ma	it	wasn	being	hadn	11
its	against	by	yourselves	through	_
o	these	how	not	because	0
what	ve	them	can	out	e
don	her	in	up	if	would
does	are	from	on	mustn	also
didn	wouldn	under	having	below	j
most	theirs	down	of	shouldn	may
same	whom	only	each	aren	r
their	s	where	y	do	10
and	you	all	nor	isn	9
did	now	haven	herself	have	l
your	as	yourself	t	yours	c
which	won	into	should	above	7
further	itself	been	she	me	1
few	needn	d	ours	my	6
to	or	such	weren	here	5
so	why	had	than	more	4
they	before	some	that	themselves	3
those	be	we	hasn		2
when	doing	ourselves	has		

risk case retrieval. Because there are still some limitations in current NLP techniques [16], some meaningless words are produced after Tokenisation, e.g. the symbol underline and the letter "j". Removal of these manually selected meaningless words with the highest numbers of occurrence could effectively reduce the size of data and this method has been adopted in some previous studies, e.g. Fan and Li [49].

- Establishing the risk case corpus: corpus in the NLP context refers to a large collection of texts [31] and this process is to combine the processed textual information into a corpus for further use in the query operation and retrieval application.

3.3. Query operation process

A basic semantic similarity problem is often observed that terms of the original query are different to the ones used in the documents in describing the same semantics [42]. To deal with the mismatching problem, a promising solution is to use query expansion [42,50,51]. In definition, query expansion is a process of reformulating or expanding a seed query using semantically related words (e.g. hyponyms, synonyms) to improve the retrieval performance of IR systems [52]. Many web IR efforts have adopted this approach and a common way is to extract the semantically related words from WordNet [41–43], a lexical database for the English language.

Because the collected risk cases are in different styles of expression by different individuals or organisations, the above problem also commonly exists in the risk case database, e.g. "structural failure" and "structure collapse". Therefore this paper integrates query expansion into the RCRS for this mismatching problem. However, WordNet is a relatively complete lexical database for the whole English environment and contains too much data which is not useful for the risk case retrieval context. For example, the synonyms of "failure" are "non-starter", "loser" and "unsuccessful person" which are not related to project risk management. In addition, no such dictionary or database has been found for defining the semantically related words in a risk management context. Hence, this paper established a small risk-related lexicon to overcome this limitation and combines the use of this risk-related lexicon and WordNet.

The pre-defined risk-related lexicon is a dictionary consisting of 107 key words, which are most commonly used in the risk management context, and their expansion suggestions. An example is shown in Fig. 6. To develop the lexicon, three major steps were used. Firstly, the 107 key words (e.g. "building", "risk", "collapse", "change", "safety") were manually selected from all risk factors in a risk database established by a previous study [53]. The second step performed a deep learning approach to find out the most related words (i.e. "Values" in Fig. 6) of 107 key words by using Word2vec [54,55], a deep learning algorithm developed by a research group led by Tomas Mikolov at Google. Word2vec is an unsupervised learning tool for obtaining vector representations for words and could be used for finding out most similar or related words in an N-dimensional vector environment. The collected 590 risk cases were initially used for training but it was quickly realised the size of data was so small that the performance of calculation is not as good as the authors expected. Then, the free and open Wikipedia content database [56] is used as a supplement for calculating the most similar words. In the third step, similar words calculated by using both risk case content corpus and Wikipedia content database are gathered together and a manual selection process based on knowledge and experience is conducted to delete words that are not related to the risk management context.

The work flow of query expansion is shown in Fig. 7. Specifically, a new query is firstly read and processed by SoA. Secondly the processed

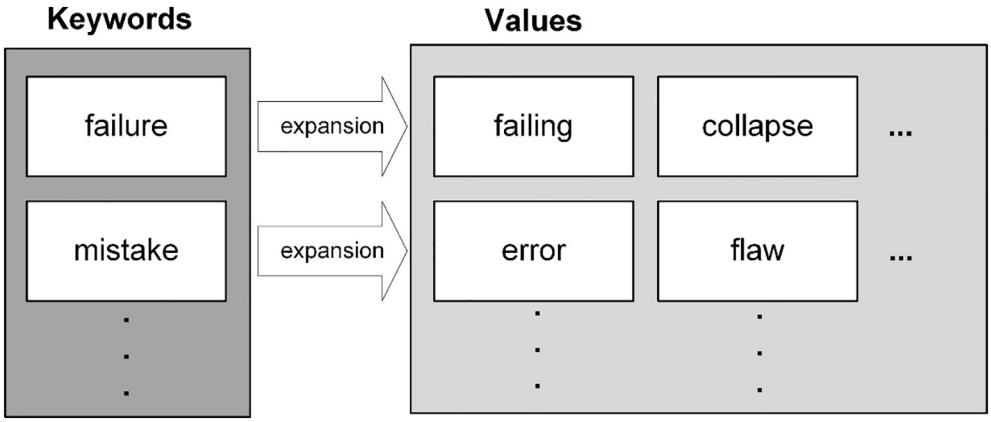

Fig. 6. Example of risk-related lexicon.

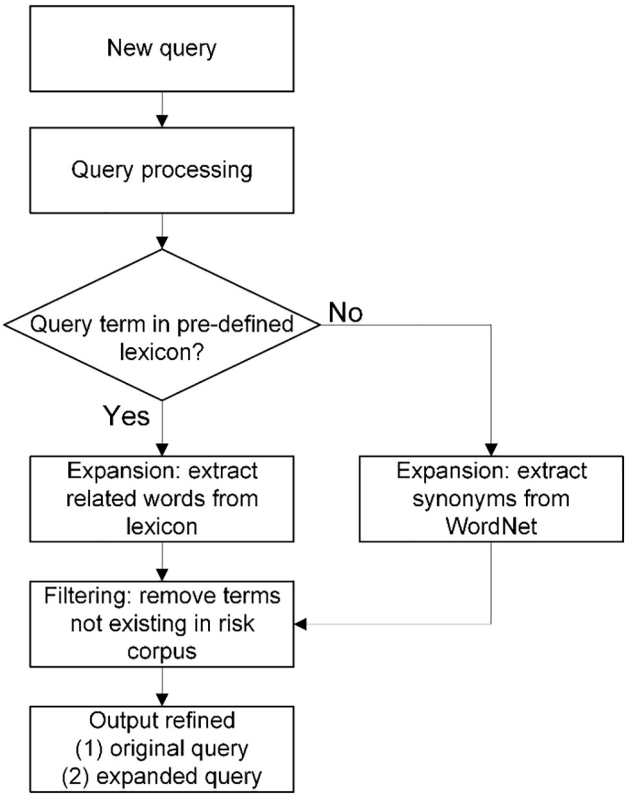

Fig. 7. Work flow of query expansion.

query terms are prior scanned to match its expansion of related words in the pre-defined risk-related lexicon. If any terms are not found in the pre-defined risk-related lexicon, they are expanded by using synonyms in WordNet. After this, there are two queries, i.e. original query, expanded query. With the observation that original query could mostly reflect a user's need for case retrieval, this paper keeps the original query and expanded query as two separate queries. Thirdly, the system scans the terms in both original query and expanded query, and removes terms that do not exist in the risk case content corpus. Lastly, the system outputs both refined original query and expanded query for further use in retrieval application.

3.4. Retrieval application process

3.4.1. The classical Vector Space Model (VSM)

In definition, the VSM is an algebraic model for representing textual documents as vectors of identifiers and assigning non-binary weights to index terms in queries and in documents, which is broadly used to compute the degree of similarity between each document and the query [32,57,58]. The classical VSM is described as follows [32]:

Query q and document d_j can be represented as t-dimensional vectors, as shown in Eqs. (1) and (2). For the vector model, t is the total number of index terms and each dimension corresponds to a separate index term. The elements $w_{i,j}$ in each vector is the weight associated with a term-document pair (k_i, d_j) and $w_{i,j} \geq 0$.

$$\vec{q} = (w_{1,q}, w_{2,q}, \ldots, w_{t,q}) \tag{1}$$

$$\vec{d_j} = (w_{1,j}, w_{2,j}, \ldots, w_{t,j}) \tag{2}$$

In the classical VSM, $w_{i,j}$ is known as the Term Frequency-Inverse Document Frequency (TF-IDF) weight. If the weight vector model for a document d_j is $\vec{d_j}$, the document's TF-IDF weights can be quantified as:

$$w_{i,j} = \left(1 + \log f_{i,j}\right) \times \log\left(\frac{N}{n_i}\right) \tag{3}$$

where $f_{i,j}$ is the frequency of index term k_i in the document, N is the total number of documents in the document set, and n_i is the number of documents containing the term k_i.

Through using the VSM and TF-IDF model, the degree of similarity $sim(d_j, q)$ between the document d_j and the query q can be quantified as the cosine of the angle between the vectors $\vec{d_j}$ and $\vec{q}$:

$$sim(d_j, q) = \frac{\vec{d_j} \cdot \vec{q}}{|\vec{d_j}| \times |\vec{q}|} = \frac{\sum_{i=1}^{t} w_{i,j} \times w_{i,q}}{\sqrt{\sum_{i=1}^{t} w_{i,j}^2} \times \sqrt{\sum_{i=1}^{t} w_{i,q}^2}} \tag{4}$$

where $|\vec{d_j}|$ and $|\vec{q}|$ are the norms of the document and query vectors, and $\vec{d_j} \cdot \vec{q}$ is the inner product of the document and query vectors.

3.4.2. The proposed score strategy and computational process

A number of existing studies [43,59] have validated that query expansion could effectively improve the IR performance and a common method for query expansion is to use WordNet or other lexical databases. WordNet has pre-defined the basic semantic relationships between words, e.g. hypernym, synonym, hyponym. Gong et al. [42,60] pointed out these different semantic relations between words for query expansion will lead to different effects on the IR performance and an easy and effective approach to distinguish their effects is to give different weighting coefficients to the expanded terms.

After considering the effect of the expanded query q_e, this study takes the classical VSM as a starting point and proposes the following method to compute the similarity between the query and risk case:

$$score = sim(d_j, q_o) + \lambda \times sim(d_j, q_e) \tag{5}$$

where λ is the coefficient for the effect of q_e and $0 < \lambda < 1$, and this study takes $\lambda = 0.7$.

The reasons are discussed as follows:

- The basic assumption of this study is that the original query and expanded query will cause different effects on the retrieval results. The original query by the user could mostly reflect a user's searching need for the risk case retrieval, and expanded terms using pre-defined risk-related lexicon or WordNet are more or less different with the original query in semantics. Therefore an optimal solution to distinguish the effects of the original query and the expanded query is to keep the original query and expanded query as separate operations (i.e. two queries q_o and q_e), and allocate different coefficients for them [42]. The expanded query q_e can be considered as an additional interpretation for the original query q_o. If the coefficient for q_o is 1, then it is clear that the coefficient for q_e should be < 1.
- As discussed in Section 3.3, this paper combines the use of a pre-defined risk-related lexicon and synonyms in WordNet as the databases for query expansion. The suggested expansion terms in the risk-related lexicon are "synonyms" of the keyword in the project risk management context. Therefore, all expanded terms can be considered similarly as "synonyms" of the original query. A previous study by Gong et al. [42] tested the performance of a web IR system using the different semantic relations between words of WordNet for query expansion, and demonstrated that the optimal value of coefficient for synonyms is 0.7. Hence this study takes λ as 0.7 for practical implementation.

The computational process is illustrated as follows. Assume there are totally k risk case documents in the risk case database, a term-document weighting matrix can be constructed as shown in Fig. 8, where the two queries are extended as the last two "documents". For each risk case or document, the TF-IDF weights of all terms are

$$\begin{bmatrix} & Doc_1 & Doc_2 & \cdots & Doc_j & \cdots & Doc_k & q_o & q_e \\ Term_1 & W_{1,1} & W_{1,2} & \cdots & W_{1,j} & \cdots & W_{1,k} & W_{1,k+1} & W_{1,k+2} \\ Term_2 & W_{2,1} & W_{2,1} & \cdots & W_{2,j} & \cdots & W_{2,k} & W_{2,k+1} & W_{2,k+2} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ Term_i & W_{i,1} & W_{i,2} & \cdots & W_{i,j} & \cdots & W_{i,k} & W_{i,k+1} & W_{i,k+2} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ Term_n & W_{n,1} & W_{n,2} & \cdots & W_{n,j} & \cdots & W_{n,k} & W_{n,k+1} & W_{n,k+2} \end{bmatrix}$$

Fig. 8. Term-document weighting matrix.

presented in a row. If a document contains no specific term, then this term's weight in the document is 0.

For any document d_j, the similarity between the query q and d_j can be computed as:

$$score = sim(d_j, q_o) + 0.7 \times sim(d_j, q_e) = \frac{\sum_{i=1}^{n} w_{i,j} \times w_{i,k+1}}{\sqrt{\sum_{i=1}^{n} w_{i,j}^2} \times \sqrt{\sum_{i=1}^{n} w_{i,k+1}^2}} + 0.7 \times \frac{\sum_{i=1}^{n} w_{i,j} \times w_{i,k+2}}{\sqrt{\sum_{i=1}^{n} w_{i,j}^2} \times \sqrt{\sum_{i=1}^{n} w_{i,k+2}^2}} \quad (6)$$

Due to the combination effects of q_o and q_e, the range of overall similarity is from 0 to 1.7.

4. System development and implementation

4.1. Prototype development

In order to fully implement the proposed RCRS, a prototype was developed using the Python programming language. Although other programming languages (e.g. R, Java) could have been used, this study chose Python because:

- Python is one of most widely used object-oriented programming languages with lots of features such as free and open source, easy syntax, and good extensibility. This means a Python program is easily read and understood by others and is highly extensible.
- A number of existing tools have been designed to support Python working with NLP, e.g. NLTK [47], data mining and analysis, e.g. scikit-learn [61]. Therefore developing the prototype using Python could build on valuable previous work and avoid repeated modelling work.

4.2. Illustrative example

The purpose of this sub-section is to use the example of "Worker Fall from Height" to illustrate the computational process of the developed prototype system. The overall computational process is presented in Fig. 9.

The overall computational process can be described as follows:

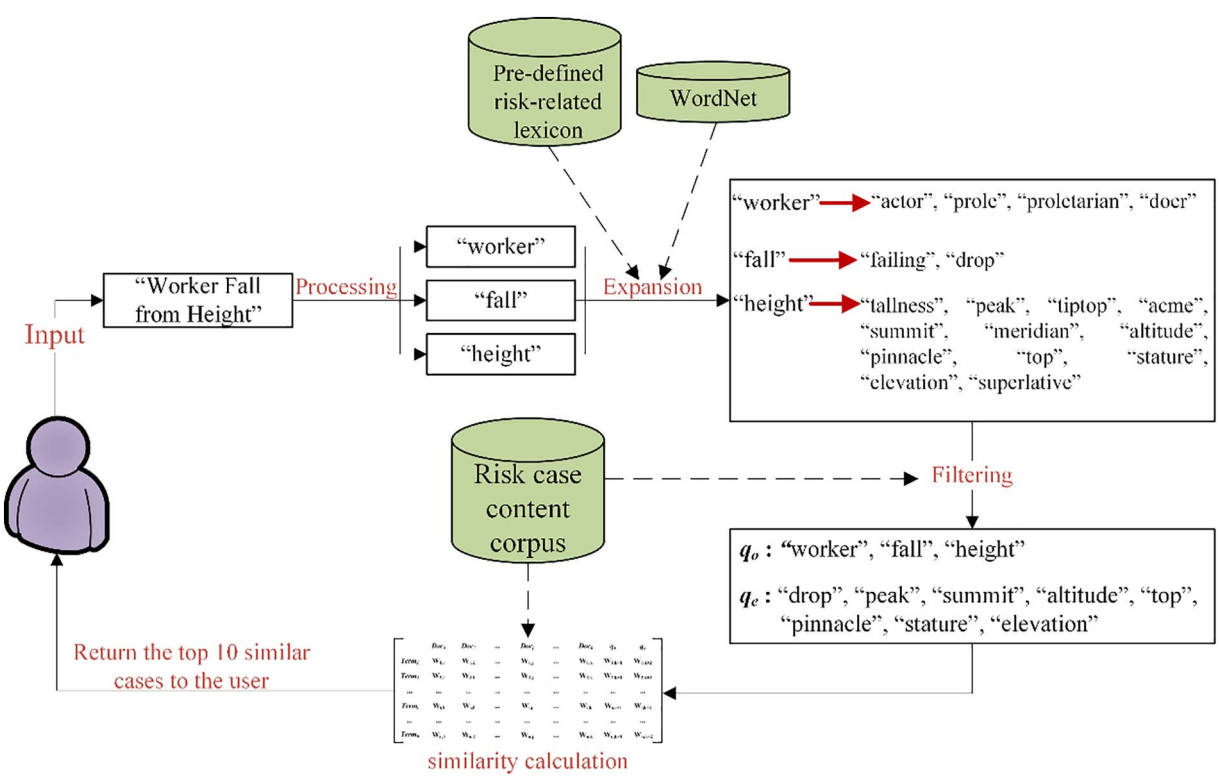

Fig. 9. Computational process of retrieving "Worker Fall from Height" similar cases.

Table 2
Top 10 similar cases of "Worker Fall from Height".

Similarity	Title of risk case	Source	Number
0.355807864882	Young worker falls from third-storey balcony	WorkSafeBC	30
0.350710609398	Fall from roof with too much slack in lifeline	WorkSafeBC	3
0.306337588766	Hispanic laborer dies after falling through a second story floor opening	NIOSH	5
0.286606375085	Worker falls through roof insulation to concrete floor	WorkSafeBC	27
0.282279911804	Worker died after fall from steep-sloped roof	WorkSafeBC	12
0.281084486537	Worker entangled in chain falling from dismantled conveyor	WorkSafeBC	13
0.278102714551	Worker died after being submerged in flooded cranberry field	WorkSafeBC	11
0.277708195414	Workers seriously burned in flash fire	WorkSafeBC	20
0.238392609973	Hispanic worker falls from residential roof	NIOSH	1
0.235168098338	Workers fall when unsecured bin tips off elevated forks	WorkSafeBC	19

- Before starting risk case retrieval, the system needs to read and process all the risk cases and establish a corpus for further use. As discussed in Section 3.2, a total of 590 risk cases have been collected. The system starts with extracting textual content from each risk case and getting the name list of all risk cases. After reading each case, the system processes its textual content through SoA, and saves the processed case in a temporary file. Then, all temporary files are read according to the sequence of name list and stored in a list where each risk case is a string.
- If a new query "Worker Fall from Height" is given by the user, the system first processes the query through SoA and obtains the tokens of original query, i.e. "worker", "fall" and "height". Then each token in the processed original query is prior scanned to find out its related words in the pre-defined lexicon. The terms not found in the pre-defined risk-related lexicon are expanded by using synonyms in WordNet. As only "fall" exists in the keyword list of pre-defined lexicon, the pre-defined lexicon is used for expansion of "fall" and the synonyms of WordNet is used for expansion of "worker" and "height". The related words for "fall" are "falling" and "drop". The related words for "worker" are "actor", "prole", "proletarian" and "doer". And the related words for "height" are "tallness", "peak", "tiptop", "acme", "summit", "meridian", "altitude", "pinnacle", "top", "stature", "elevation" and "superlative". Thirdly, the system filters the original query and expanded query by scanning the risk case content corpus and deleting those terms that do not appear in the corpus. After filtering, the original query are "worker", "fall" and "height" and the expanded terms are "drop", "peak", "summit", "altitude", "top", "pinnacle", "stature" and "elevation".
- In the third step, the processed original query and expanded query are first extended to the corpus as the last two strings in the list. Then the system performs the calculation of TF-IDF weights and establishes the corresponding term-document matrix (shown in Fig. 8). Finally, the similarity between the query and each risk case is computed by using Eq. (6) and the system returns the ranked top 10 similar risk cases to the end users. The result is shown in Table 2.

4.3. System testing

Although there are a number of matrices that have been proposed to evaluate and test IR systems, the most widely used are Precision, Recall and F score [14,16,32] which can be calculated with the help of a simplified confusion matrix [32,62] shown in Table 3. There are four variables in the simplified confusion matrix, i.e. True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN). Here the terms "positive" and "negative" mean the expectation of a retrieval while the terms "true" and "false" refer to whether that expectation corresponds to the external judgment. In other words, TP means the number of relevant documents retrieved, FP means the number of irrelevant documents retrieved, FN means the number of relevant documents not retrieved, and TN means the number of irrelevant documents not retrieved.

Table 3
Confusion matrix.

	Relevant	Not relevant
Retrieved	True Positive (TP)	False Positive (FP)
Not retrieved	False Negative (FN)	True Negative (TN)

Precision refers to the fraction of retrieved documents that is relevant and is used to measure the percentage of relevant documents in all retrieved documents, i.e.

$$Precision = \frac{TP}{TP + FP} \times 100\% \tag{7}$$

Recall is defined as the fraction of relevant documents that has been retrieved and used for measuring the percentage of retrieved documents in all relevant documents, i.e.

$$Recall = \frac{TP}{TP + FN} \times 100\% \tag{8}$$

Another measure called F is the harmonic mean of Precision and Recall and is defined as follows:

$$F = \frac{Precision \times Recall}{Precision + Recall} \times 100\% \tag{9}$$

It is noticed that Precision, Recall, and F value are commonly used for evaluating the whole retrieval system and it requires an accurate boundary between "retrieved" and "not retrieved" to calculate the three measures. Here determining the threshold (or cut-off) is extremely important and its value could in large degree affect the evaluation results of an IR system. However, there is a need to point out that determining the threshold value in an IR system is complex and needs a large number of experiments, which is not within the scope of this study. Unlike web-scale IR, the information in the construction industry is relatively small-scale and domain-specific and a common method to evaluate the performance of an IR system for construction projects is through testing a number of samples and setting user experience based threshold value, e.g. [16,49]. Besides, with the observation that in the real working environment engineers often expect to obtain the needed information within a limited amount of time [63] and the top 10–20 cases would by nature have the most value to the end users [49], the proposed RCRS is designed to return the top 10 most similar cases. Hence, this study also evaluated the percentage of relevant risk cases among the top 10 similar documents, which is defined as Precision at 10 (P@10):

$$P@10 = \frac{number\ of\ relevant\ documents\ in\ top\ 10}{10} \times 100\% \tag{10}$$

In order to test and evaluate the proposed RCRS, this study took the threshold value as 0.1 from preliminary system use experience and the testing procedure consists of the following steps:

- Firstly, a set of key terms (e.g. "bridge", "fall", "collapse", "con-

Table 4
Testing results.

No.	Testing query	Number of retrievals				Performance			
		TP	FP	FN	TN	Precision	Recall	F	P@10
1	fall from height	18	1	18	553	94.7%	50.0%	65.5%	90%
2	flood risk	11	5	0	574	68.8%	100.0%	81.5%	100%
3	design error	22	4	6	558	84.6%	78.6%	81.5%	100%
4	flood risk of bridge	11	30	0	549	26.8%	100.0%	42.3%	100%
5	worker fall from height	25	10	2	553	71.4%	92.6%	80.6%	90%
6	tower crane collapse	18	23	0	549	43.9%	100.0%	61.0%	70%
7	bridge failure	42	16	3	529	72.4%	93.3%	81.6%	100%
8	worker injury	32	3	18	537	91.4%	64.0%	75.3%	100%
9	worker die in construction	30	1	11	548	96.8%	73.2%	83.3%	100%
10	structure collapse in demolition	16	34	0	540	32.0%	100.0%	48.5%	100%

struction") that are relevant to the scope of collected risk cases were selected for making up 10 testing queries. The queries were divided into 3 groups, i.e. "type of risk", "object + type of risk", and "object + type of risk + project phase", to simulate the real situations of case retrieval. The "type of risk" group contains three queries, i.e. "fall from height", "flood risk", "design error". The "object + type of risk" group consists of 5 queries, i.e. "flood risk of bridge", "worker fall from height", "tower crane collapse", "bridge failure", "worker injury". The "object + type of risk + project phase" group contains two queries, i.e. "worker die in construction" and "structure collapse in demolition";

- Secondly, each testing query was inputted into the RCRS for query-document matching and the corresponding output was recorded in an Excel table. As this paper took an experience-based threshold (or cut-off) value 0.1, those documents with the similarity score over 0.1 were classified into the "retrieved" group while those documents with the similarity score which is < 0.1 were classified to the "not retrieved" group;
- Thirdly, because the similarity value for those documents containing no terms of original and expanded queries is 0, then those documents were determined to be irrelevant directly. Then the results were carefully reviewed to determine if a risk case is relevant to the query by quickly reading and understanding each document and analysing the relationship between the query and the document. If a document is determined to be relevant to the query, the value "1" was labelled for that document in Excel. Otherwise, the value "0" was given. Then, TP, FP, FN, TN and P@10 were calculated.
- In the last step, the calculation of Precision, Recall, and F value for each testing retrieval was performed and the testing results are shown in Table 4.

The search results show that generally the proposed RCRS is capable of retrieving relevant risk cases from the database for a specified query. In particular, the results of P@10 are excellent, mostly 100% (7 of 10). Only one testing query had 70% of P@10, which also is a satisfactory result. Therefore the top 10 cases returned by the system are valuable to the user. The high percentage of P@10 can be explained by the term frequency being an important factor in computing the TF-IDF weights and a document containing as many query terms as possible is easier to obtain a high similarity score. Although the Precision score for several queries were relatively low, this does not mean the retrieval results were not good. For example, for the "flood risk of bridge" query, 41 results were retrieved and only 11 were determined to be similar to the query. Two reasons could explain this problem: first, there are a very small number of "flood" related samples in the risk case database; second, because the threshold value 0.1 in this case is too small and the expanded terms were producing some "noise". But from its P@10 score, it can be seen that the top 10 were all similar to the query and nearly all valuable documents were ranked. Therefore simply increasing the threshold value for some queries could improve the search results. In addition, some researchers [14,16] also claim that there are still some technical limitations in the current NLP, which lead to the conclusion that the search results cannot be perfect. For example, the "flood risk" here is an entity but the system failed to read it as an entity and split it into two separate terms "flood" and "risk" for consideration.

5. Discussions

The literature shows that CBR is a process of learning from the past, which could facilitate previous knowledge and experience to be effectively used for risk management in new projects. In the CBR cycle, RETRIEVE is the first and the most important step [7,15]. A commonly used traditional way for assessing the similarity between user need and risk cases is through attaching attribute labels to each risk case document and allocating different weights to those attributes [9,22,25]. However, as discussed in Section 2.1, some challenges still exist: (1) traditional methods are very limited in scope, (2) a large amount of pre-processing or preparation work is needed, and (3) very few studies have been found to be capable of addressing the challenge of semantic similarity. In order to overcome the current challenges of case retrieval in CBR, this paper analysed the potential and benefits of integrating NLP into risk case retrieval. The idea was motivated by recent research that has introduced NLP into textual information management into construction industry, e.g. retrieval of CAD drawings [16], retrieval of relevant information for assisting decision making [64,65], injury report content analysis [14], and document clustering [17]. It can be seen that the application of NLP into textual documents analysis and management in the construction industry is a new and promising trend. Some recent studies even extended the use of NLP into Building Information Modelling (BIM), an emerging digital technology in the construction industry, for automated code checking [66], processing building information [67], retrieving online BIM resources [50], etc.

A number of recent studies [16,49] successfully used the classical VSM for IR and document management, and discussed that the semantic similarity is still a huge challenge in any current application of NLP in the construction industry. To partially overcome this gap, this paper outlines a framework of combining the use of semantic query expansion and VSM for retrieval of similar risk cases, and develops a system prototype with Python to support the proposed approach. The test results show the proposed system could quickly and effectively retrieve and rank valuable risk cases when a query is specified. Through implementing the proposed system, end users could quickly find out risk cases that are valuable references to the new situations or problems and embed the knowledge and experience of previous accidents into daily work. Any new cases could be added into the risk case database flexibly for retrieval without pre-processing work. In addition, because this system prototype is written with Python, the RCRS could also be easily integrated into software written by other programming languages. As an example of its practical contributions, the proposed

approach can be embedded into some online risk case databases, e.g. Structural-Safety and NIOSH, as a semantic searching engine. In the future, the proposed approach can be also expanded for the wider management of engineering documents and information.

Of course, some limitations also exist in this study. These limitations and the corresponding recommendations for future research are discussed as follows:

- First, the proposed system is limited in case retrieval within the internal risk case database and the total number of collected risk cases is still relatively small. As described in Section 3.2, due to the limited time only 590 risk cases covering 7 types of risk were collected. The reasons are: 1) the main purpose of this study is developing a general approach (i.e. proof of concept) based on NLP for risk case retrieval instead of establishing a complete risk case database; and 2) there are relatively few detailed reports on those risks that are not so dangerous or fatal, e.g. financial loss, time overrun. However, the limited size of the database will influence the retrieval results and practical applicability. For example, if a user query is "time overrun" and the database contains no risk cases about "time overrun", it will be difficult for the system to return the desired results to the user. Therefore, future research may consider: 1) how to enrich the risk case database; 2) how to formulate case retrieval guidelines to the end user according to the distribution of risk cases; and 3) how to extend the proposed system for risk case retrieval in external databases and online resources.
- Secondly, the semantic similarity problem is still a huge challenge within the state-of-the-art research of NLP [31], and the query expansion approach adopted by this study can only address a limited proportion of the problem. In particular, the proposed system combines the use of a pre-defined risk-related lexicon and WordNet to deal with the word mismatching problem of case retrieval. However, the pre-defined lexicon only contains explanations of 107 key terms in the project risk management domain and is not a complete dictionary. To overcome the shortcoming of the pre-defined lexicon, WordNet is used as an important supplementary. However, because WordNet is a large lexical database for the English language and is not specially designed for risk management, this study found some terms expanded by WordNet are not related to project risks and have little, or no value in risk case retrieval. Moreover, it can be seen that human language is still extremely complex and difficult for computers to understand and process. For example, Caldas and Han [68] made use of IR and text mining for automatic classification of project documents but found the results were not perfect due to the multiple meanings of words. In addition, as discussed in Section 4.3, though the pre-defined lexicon and WordNet can be used for explanation of a single term, it is still difficult for computer to process the word groups. Hence, one short-term recommendation for future research may be to establish a comprehensive lexicon for project risk management which includes the definition of the linked relationships of common word groups. From a long-term perspective, future research may apply the state-of-the-art techniques of NLP into addressing the semantic similarity problem in both risk case retrieval and other fields.
- Thirdly, the proposed system has not been put into use and validated in practice. For better implementation of the proposed approach, the prototype system needs to be further developed as a tool with easy-to-use user interface and checked by different scenarios. In addition, as the proposed system was designed to return the most similar 10 risk cases to the user and the test results presented in Sections 4.2 and 4.3 are satisfactory, when conducting the preliminary testing this paper checked the results manually and did not study the best value of the threshold. Although a number of matrices (e.g. Precision, Recall, F and P@10) could be used for evaluating an IR system, nearly all of them require a clear boundary of "retrieved" and "not retrieved", and "relevant" and "not relevance". The threshold value is often used to divide the returned results into "retrieved" and "not retrieved"; however, Qady and Kandil [17] pointed out the best threshold value normally lies between 0.05 and 0.95, and determining the best value needs a large number of experiments. Furthermore, the relevance is by nature often continuous instead of binary, which leads to the difficulty of determining if a retrieved document is relevant or not [69,70]. Hence, future research may further study the threshold value and relevance problem, and test and improve the proposed approach and system in real practice.

6. Conclusions

This paper introduced an approach of combining the use of two NLP techniques (i.e. VSM and semantic query expansion) for risk case retrieval and proposed a framework for the Risk Case Retrieval System. The VSM could represent textual documents as vectors of identifiers and assigning TF-IDF weights to index terms in both queries and documents, which could be used to compute the degree of similarity between documents and the query, while the query expansion could solve the mismatching problem of terms that have the same semantic meanings through expanding the original query using related terms defined in a pre-defined risk-related lexicon and synonyms in WordNet. A prototype system was developed using Python to implement the proposed approach.

Through implementing the proposed system, textual content information is firstly extracted from the risk case dataset and processed to generate a content corpus. After a query is inputted by the user, then the system starts to read and process the query, combines the use of a pre-defined risk-related lexicon or WordNet to expand the original query, and filters out the query terms that do not exist in the content corpus. Lastly the system gathers original query, expanded query and content corpus together for query-document similarity computing and returns the top 10 similar risk cases to the user. The preliminary test results have demonstrated the system's capacity of automatically retrieving similar risk cases.

Although there are still some limitations of applying current NLP technology into engineering textual information management, using such a system for managing risk cases could effectively facilitate the risk identification and communication, and information management. The suggested future research may include, for example: 1) to enrich the risk case database and expand the capacity of the proposed system for accessing both internal database and online risk case resources; 2) to investigate how state-of-the-art NLP can be further developed to address the semantic similarity problems (e.g. processing word groups); 3) to improve the evaluation methods for retrieval of small-scale data; and 4) to test and optimise the proposed approach and system in practice.

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